Cartesian Products

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Exercises

1. Show there is a bijective correspondence of $A \times B$ and $B \times A$.

Proof. It is claimed that the function $f : A \times B \to B \times A$ defined as $f((a, b)) = (b, a)$. To show f is surjective, consider $(b, a) \in B \times A$. Then, by definition $b \in B$ and $a \in A$. As such, $(a, b) \in A \times B$, and therefore $f((a, b)) = (b, a)$. Thus, since (b, a) was an arbitrary element of $B \times A$ it has been shown that for any element in $s \in B \times A$ there exists at least one element $t \in A \times B$ such that $f(t) = s$. Hence, f is surjective.

Consider now $(a, b), (c, d) \in A \times B$ such that $f((a, b)) = f((c, d)) = (s, t)$. By definition of f, note $a = t$ and $c = t$ implying $a = c$. Similarly, it must be that $b = d$, and thus $(a, b) = (c, d)$. Therefore, f is injective, and it can be thus concluded that f is bijective. □

2.

(a) Show that if $n > 1$ there is a bijective correspondence of

$$A_1 \times \cdots \times A_n \quad \text{with} \quad (A_1 \times \cdots \times A_{n-1}) \times A_n.$$

Proof. If $n = 2$, note $A_1 \times A_2 = (A_1) \times A_2$ and thus there exists a bijective correspondence between both sets, in particular this bijective map is the identity map. Thus, assume $n > 2$. We claim that $f : A_1 \times \cdots \times A_n \to (A_1 \times \cdots \times A_{n-1}) \times A_n$ defined by $(a_1, \ldots, a_n) \mapsto ((a_1, \ldots, a_{n-1}), a_n)$ is bijective. We begin by showing f is injective. Let $(a_1, \ldots, a_n)$ and $(b_1, \ldots, b_n)$ be elements in the domain set such that they both map to $((c_1, \ldots, c_{n-1}), c_n)$. By construction, $a_n = c_n$ and $b_n = c_n$ implying $a_n = b_n$. Similarly, $a_1, \ldots, a_{n-1}$ must all equal the corresponding element in $(c_1, \ldots, c_{n-1})$, i.e., $a_1 = c_1$ and so on. In the same way, $b_i = c_i$ for $1 \leq i \leq n - 1$. Therefore, $a_i = b_i$ for $1 \leq i \leq n-1$, and we thus conclude $(a_1, \ldots, a_n) = (b_1, \ldots, b_n)$ making f injective.
To show f is surjective, consider the element $((c_1, \ldots, c_{n-1}), c_n)$ in the set being mapped onto by f. By construction, $c_i \in A_i$ for $1 \leq i \leq n$. Thus, $(c_1, \ldots, c_n) \in A_1 \times \cdots \times A_n$. By definition of f, $f(c_1, \ldots, c_n) = ((c_1, \ldots, c_{n-1}), c_n)$. Thus, we have shown f is also surjective, and therefore we conclude f is bijective. □

(b) Given the indexed family $\{A_1, A_2, \ldots\}$ let $B_i = A_{2i-1} \times A_{2i}$ for each positive integer i. Show there is a bijective correspondence of $A_1 \times A_2 \times \cdots$ with $B_1 \times B_2 \times \cdots$.

Proof. Note $B_1 \times B_2 \times \cdots$ is equivalent to the set $(A_1 \times A_2) \times (A_3 \times A_4) \times \cdots$ by construction. Let $A = A_1 \times A_2 \times \cdots$ and $B = (A_1 \times A_2) \times (A_3 \times A_4) \times \cdots$. Consider then the natural map $f : A \to B$ defined by $(a_1, a_2, a_3, a_4, \ldots) \mapsto ((a_1, a_2), (a_3, a_4), \ldots)$. We claim that f is a

bijective correspondence between the two sets. Let $(s_1, s_2, ...), (t_1, t_2, ...) \in A$ such that f maps both of these elements of A onto $((m_1, m_2), ...) \in B$. Then, by definition of f, $s_n = m_n$ and $t_n = m_n$ for all $n \in \mathbb{Z}_+$. Thus, $s_n = t_n$ for all n which implies $(s_1, s_2, ...) = (t_1, t_2, ...)$. As such, f is an injective function. Consider then $((p_1, p_2), (p_3, p_4), ...)$. By definition of B, $p_n \in A_n$ for some $A_n \in \{A_1, A_2, ...\}$. As such, $(p_1, p_2, p_3, p_4, ...) \in A$, and thus f maps this element of A onto $((p_1, p_2), (p_3, p_4), ...)$. Therefore, f is surjective, and it can be concluded that f is a bijective correspondence between A and B. □

3. Let $A = A_1 \times A_2 \times \cdots$ and $B = B_1 \times B_2 \times \cdots$.

(a) Show that if $B_i \subset A_i$ for all i, then $B \subset A$. (Strictly speaking, if we are given a function mapping the index set $\mathbb{Z}_+$ into the union of the sets B_i, we must change its range before it can be considered as a function mapping $\mathbb{Z}_+$ into the union of the sets A_i. We shall ignore this technicality when dealing with Cartesian products).

Proof. If B is empty, then it is clearly a subset of A given that the empty set is a subset of all sets. As such, assume B is non-empty and let $(b_1, b_2, ...) \in B$. By assumption, $b_i \in B_i \subset A_i$. Thus, each $b_i \in A_i$ for all $i \in \mathbb{Z}_+$. Therefore, $(b_1, b_2, ...) \in A$ implying that $B \subset A$. □

(b) Show the converse of (a) holds if B is non-empty.

Proof. Let $B \subset A$ such that B is non-empty. Since B is non-empty, there exists some element $(b_1, b_2, ...) \in B$. In particular, each B_i is non-empty since $b_i \in B_i$ where b_i is from the tuple element guaranteed to be in B. Thus, fix some index k and let $b_k \in B_k$ be an arbitrary element of B_k. Consider then the tuple with the kth coordinate swapped, $(b_1, \dots, b_k, ...) \in B$. Since $B \subset A$, $(b_1, \dots, b_k, ...) \in A$. Therefore, $b_k \in A_k$. Thus, $B_k \subset A_k$, and since this was for arbitrary B_k we conclude that $B_i \subset A_i$ for all $i \in \mathbb{Z}_+$. □

(c) Show that if A is non-empty, each A_i is non-empty. Does the converse hold?

Proof. Let $A = A_1 \times A_2 \times \cdots$ be non-empty. Thus, there exists an element $(a_1, a_2, a_3, ...) \in A$ such that $a_i \in A_i$. Hence, A_i is not empty for all $i \in \mathbb{Z}_+$ since $a_i \in A_i$.
The converse does hold also. Let A_i be non-empty for each i, then consider $a_1 \in A_1$, $a_2 \in A_2$, and so on where each a_i is an arbitrary element of A_i. Then, by definition of A, $(a_1, a_2, a_3, ...) \in A$ which implies that A is non-empty. [This makes use of the axiom of choice informally through the "and so on"]. □

(d) What is the relation between the set $A \cup B$ and the Cartesian product of the sets $A_i \cup B_i$? What is the relation between the set $A \cap B$ and the Cartesian product of the sets $A_i \cap B_i$?

Proof. To some degree. the following statements seem correct:
$A \cup B$ is a subset of $\prod_{i \in \mathbb{Z}_+} A_i \cup B_i$, whereas $A \cap B$ is equivalent to $\prod_{i \in \mathbb{Z}_+} A_i \cap B_i$. □

4. Let $n, m \in \mathbb{Z}_+$. Let $X \neq \emptyset$.

(a) If $m \leq n$, find and injective map $f : X^m \to X^n$.

Proof. Define $f : X^m \to X^n$ by the map $(x_1, \dots, x_m) \mapsto (x_1, \dots, x_m, 0, \dots, 0)$ where the remaining trailing zeros are exactly $n - m$. Thus, if $n - m = 0$ there are no trailing zeros, and if $n - m = 3$ there are three trailing zeros. To show that f is injective let $(p_1, \dots, p_m)$ and $(t_1, \dots, t_m)$ be elements of X^m such that both are mapped by f onto the element $(s_1, \dots, s_m, 0, \dots, 0) \in X^n$. By definition of f, $p_i = s_i$ and $t_i = s_i$ for $1 \leq i \leq m$. Thus, $p_i = t_i$ for $1 \leq i \leq m$. Hence, $(p_1, \dots p_m) = (t_1, \dots, t_m)$. Therefore, f is injective. □

(b) Find a bijective map $g : X^m \times X^n \to X^{m+n}$.

Proof. Define $g : X^m \times X^n \to X^{m+n}$ to be the map such that $((p_1, \dots, p_m), (s_1, \dots, s_n))$ is mapped to $(p_1, \dots, p_m, s_1, \dots, s_n)$. To show that it is surjective, consider the element $(y_1, \dots, y_m, r_1, \dots, r_n)$ of X^{m+n}. Note $(y_1, \dots, y_m) \in X^m$ and $(r_1, \dots, r_n) \in X^n$, as such the element $((y_1, \dots, y_m), (r_1, \dots, r_n))$ is in $X^m \times X^n$, and g maps this element to $(y_1, \dots, y_m, r_1, \dots, r_n)$ by construction. Therefore, g is surjective.
Now, consider two different elements of $X^m \times X^n$ such that they are both mapped to the same element of X^{m+n} under g. Due to how g was constructed, note that the elements of $X^m \times X^n$ must be equal, and thus g is injective also. □

(c) Find an injective map $h : X^n \to X^\omega$.

Proof. Define $h : X^n \to X^\omega$ by the map $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, 0, \dots)$. Note h must be injective since if $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ are both mapped to the same element in X^ω then $y_i = x_i$ for $1 \leq i \leq n$. Therefore, h is injective. □

(d) Find a bijective map $k : X^n \times X^\omega \to X^\omega$.

Proof. Define $k : X^n \times X^\omega$ by the map $((p_1, \dots, p_n), (x_1, \dots)) \mapsto (p_1, \dots, p_n, x_1, \dots)$. To show that k is surjective, consider the element $(s_1, \dots, s_n, s_{n+1}, \dots) \in X^\omega$, and note $(s_1, \dots, s_n)$ is an element of X^n. Thus, $((s_1, \dots, s_n), (s_{n+1}, \dots))$ is an element of $X^n \times X^\omega$ that is mapped onto $(s_1, \dots, s_n, s_{n+1}, \dots)$. Thus, k is surjective.
Now, assume that $((t_1, \dots, t_n), (x_1, \dots))$ and $((v_1, \dots, v_n), (x_1, \dots))$ are two elements of $X^n \times X^\omega$ such that they are both mapped to $(k_1, \dots, k_n, x_1, \dots) \in X^\omega$. Then, by construction of k, $t_i = k_i$ and $v_i = k_i$ for $1 \leq i \leq k$, and x_i are all equal. In particular, this implies $(t_1, \dots, t_n) = (v_1, \dots, v_n)$, and thus the two elements of $X^n \times X^\omega$ are also equal. Therefore, k is also injective, and as such we conclude k is bijective. □

(e) Find a bijective map $l : X^\omega \times X^\omega \to X^\omega$.

Proof. Define $l : X^\omega \times X^\omega \to X^\omega$ by the map $((x_1, x_2, \dots), (y_1, y_2, \dots)) \mapsto (x_1, y_1, x_2, y_2, \dots)$. To show that l is surjective, consider the element $(s_1, s_2, \dots) \in X^\omega$ and note that $s_{2i-1} \in X$ and $s_{2i} \in X$ for all $i \in \mathbb{Z}_+$. Thus, $(s_1, s_3, \dots) \in X^\omega$ and $(s_2, s_4) \in X^\omega$. Hence, we note that $((s_1, s_3, \dots), (s_2, s_4))$ is an element of $X^\omega \times X^\omega$ that is mapped onto $(s_1, s_2, \dots)$ by k. Thus, k is surjective.
To show k is injective, consider two elements in $X^\omega \times X^\omega$, say x, y, such that they are both mapped to the same element of X^ω, z. This would imply that the elements in the first tuple of x and y are the odd elements in z, and the elements of the second tuple of x and y are the even elements of z. As such, element wise each tuple of x and y are equal, which implies x and y must be the same element of $X^\omega \times X^\omega$ since they are element wise the same. As such, l is also injective, and thus we conclude l is bijective. □

(f) If $A \subset B$, find an injective map $m : (A^\omega)^n \to B^\omega$.

Proof. Define $m : (A^\omega)^n \to B^\omega$ by the map $(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) \mapsto (\mathbf{x}_{11}, \mathbf{x}_{21}, \ldots, \mathbf{x}_{n1}, \mathbf{x}_{12}, \ldots)$. Where $\mathbf{x}_{ij}$ refers to the jth element of the ith tuple. To show that this map is injective, consider the elements $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ and $(\mathbf{y}_1, \ldots, \mathbf{y}_n)$ of $(A^\omega)^n$ such that they both are mapped by m onto the element $(b_1, b_2, \ldots)$ of B^ω. Note $\mathbf{x}_{ij}, \mathbf{y}_{ij} \in B$, and thus by construction of m, it must be that each element of $\mathbf{x}$ and $\mathbf{y}$ must be the same, implying that these sequence elements of $(A^\omega)^n$ are element wise equal, which is the same as them being equal. Therefore, m is injective. □

5. Which of the following subsets of $\mathbb{R}^\omega$ can be expressed as the Cartesian product of subsets of $\mathbb{R}$?

(a) $\{\mathbf{x} \mid x_i \text{ is an integer for all } i\}$.

Proof. This subset of $\mathbb{R}^\omega$ can be expressed as the Cartesian product of subsets of $\mathbb{R}$. In particular, it can be expressed as $\mathbb{Z}^\omega$. □

(b) $\{\mathbf{x} \mid x_i \geq i \text{ for all } i\}$.

Proof. This subset of $\mathbb{R}^\omega$ cannot be expressed as the Cartesian product of subsets of $\mathbb{R}$. □

(c) $\{\mathbf{x} \mid x_i \text{ is an integer for all } i \geq 100\}$.

Proof. This subset of $\mathbb{R}^\omega$ can be expressed as the Cartesian product of subsets of $\mathbb{R}$. In particular, it could be expressed as $\mathbb{R} \times \cdots \times \mathbb{R} \times \mathbb{Z} \times \mathbb{Z} \times \cdots$, where the first 99 subsets are $\mathbb{R}$ itself, and the rest will be $\mathbb{Z}$. □

(d) $\{\mathbf{x} \mid x_2 = x_3\}$.

Proof. This subset of $\mathbb{R}^\omega$ cannot be expressed as the Cartesian product of subsets of $\mathbb{R}$. □